

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)		[It has] magnitude (accept size) and direction	1			1		
	(b)	(i)	Total momentum before collision = $30\,000 + 15\,000$ [= $45\,000\text{ kg m s}^{-1}$] (1) Total momentum after collision = $27\,000 + 18\,000$ [= $45\,000\text{ kg m s}^{-1}$] (1) Ignore units Deduct 1 mark for powers of 10 slip Award 1 mark only for - Momentum is not lost [in collision] or momentum before [collision] is the same as momentum after [collision] / momentum is conserved (1) Don't accept: they are the same Alternative: Loss in momentum of A = $12\,000\text{ kg m s}^{-1}$ (1) Gain in momentum of B = $12\,000\text{ kg m s}^{-1}$ (1) Hence the gain in momentum of B = loss in momentum of A (1)	1	1 1		3	2	
		(ii)	Attempt at using $p_A + p_B = (m_A + m_B)v$ (1) Correct substitution e.g. $45\,000\text{ ecf} = 25\,000v$ (1) Award 2 marks if this seen. $v = 1.8\text{ m s}^{-1}$ (1)	1	1 1		3	3	
	(c)	(i)	A body's <u>rate of</u> / change per second (reference to time) change of momentum (1) is proportional to [accept 'equal to'] the [resultant] force acting on it (1) [and is in the direction of this force] Alternative: Formula stated (1) with all terms defined (1)	2			2		
		(ii)	Time for collision = 0.2 s (1) accept $(0.5 - 0.3)$ $F = \frac{-12\,000}{0.2}$ (1) $F = -60\,000\text{ [N]}$ (1) ecf on powers of 10 slip		1 1	1	3	3	

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		(iii)	Newton's 3 rd Law (1) Accept N3 Law Change of momentum is +12 000 kg m s ⁻¹ or the same and collision time is the same (1) Accept: [magnitude] of gradient same Don't accept graph is symmetrical	1	1		2		
			Question 7 total	6	7	1	14	8	0